Multi-Caption Guided CLIP Adaptation for Remote Sensing Scene Classification

Project for HKUST ELEC4240: Deep Learning in Computer Vision (Fall 2025-26)

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Abstract

Remote sensing image classification is crucial for applications like land use monitoring and disaster management, but traditional methods require large labeled datasets that are costly to obtain. This project adapts the Contrastive Language-Image Pretraining (CLIP) model for zero-shot and few-shot classification on the Remote Sensing Image Captioning Dataset (RSICD). Our approach combines text augmentation, parameter-efficient fine-tuning (LoRA), a dual vision encoder (Swin Transformer + ViT), and MaPLe-style multimodal prompt learning to enhance vision-language alignment.

*Key innovations include systematic caption pairing strategies, stochastic text augmentation, and LoRA-based adaptation to minimize trainable parameters. Experiments compare multi-caption handling techniques, LoRA configurations, and dual-encoder fusion. Our best model achieves **91.9% top-1 accuracy**, outperforming the CLIP-rsicd benchmark by **2.9%** and zero-shot CLIP by **88.2%**. Multi-caption pairing with text augmentation improves accuracy by **5.1%**, while vision-only LoRA proves most effective, highlighting the dominance of visual domain shifts. The dual encoder design further enhances performance, though full integration with LoRA presents optimization challenges. MaPLe improves generalization but requires extended training for optimal results.*

These findings demonstrate that strategic caption utilization and targeted parameter-efficient tuning can significantly enhance CLIP's performance for remote sensing tasks with limited labeled data. The framework provides a scalable solution for domain adaptation while maintaining computational efficiency.

1. Introduction

Remote sensing image classification plays a pivotal role in environmental monitoring, urban planning, and disaster response by enabling automated analysis of satellite and aerial imagery. However, conventional supervised approaches [11] face two fundamental challenges: (1) the scarcity of labeled training data in specialized domains, and (2) the significant domain gap between natural images (used for pretraining vision models) and remote sensing scenes characterized by unique spatial, spectral, and geometric properties [3].

The emergence of vision-language foundation models like CLIP [9] has revolutionized zero-shot transfer learning through aligned image-text representations. While promising, CLIP's performance degrades on remote sensing tasks due to:

- Textural and semantic disparities between natural and aerial imagery
- Limited adaptation to fine-grained scene categories (e.g., distinguishing `mediumresidential` from `denseresidential`)
- Ineffective utilization of multi-caption annotations in datasets like RSICD [7]

This work addresses these limitations through three key innovations:

1. **Dual Vision Encoder Architecture**: We introduce a hybrid visual backbone combining a Vision Transformer (ViT) for global feature extraction and a Swin Transformer for hierarchical local-global representation learning. The encoders are fused via learnable weighting ($\alpha = \sigma(w)$), dynamically balancing their contributions. This design captures both high-level semantics (ViT) and fine-grained spatial patterns (Swin), achieving **87.2% top-1 accuracy** without LoRA—a **0.8% improvement** over single-encoder baselines.
2. **Parameter-Efficient Adaptation**: We optimize the dual encoder with Low-Rank Adaptation (LoRA), restricting updates to rank-decomposed matrices in attention layers. Vision-only LoRA tuning achieves **91.9% top-1 accuracy**, demonstrating that visual domain gaps dominate in remote sensing. Our ablation shows LoRA on the ViT branch alone suffices, as Swin's pretrained features require minimal adaptation.
3. **Multimodal Prompt Learning**: We integrate MaPLe-style prompting with the dual vision encoder architec-

ture, where text-derived prompts guide both ViT and Swin branches via lightweight projection layers. We identify the need for extended training (50+ epochs) to stabilize prompt convergence in the dual-encoder setting.

2. Related Work

The Contrastive Language-Image Pre-training (CLIP) model [9] has become a cornerstone vision-language foundation model, enabling zero-shot transfer to downstream tasks by aligning image and text embeddings in a shared latent space. In remote sensing (RS) image classification, datasets such as RSICD provide multiple captions per aerial image, posing unique challenges for domain adaptation. Recent works have explored efficient fine-tuning strategies, multi-caption utilization, parameter-efficient adaptation, and prompt learning—directly informing our approach to classification on the RSICD test set.

2.1. Caption Processing Techniques

RSICD and similar datasets (e.g., UCM-Captions, Sydney-Captions) contain 5 captions per image, capturing diverse semantic descriptions. Several strategies have been proposed to exploit this redundancy. Pal et al. [8] apply offline text augmentation via back-translation with MarianMT models to generate paraphrased captions, improving robustness to linguistic variation. Common pairing strategies include *single random caption* selection per batch (used in the open-source CLIP-rsicd baseline [1]), *multi-caption averaging* of text embeddings, and *concatenation* of all captions into a long prompt (explored in Long-CLIP variants for RS [10]). RemoteCLIP [2] further introduces box-to-caption (B2C) synthesis to create paired data from object-detection annotations. Our work systematically compares these pairing techniques (random selection, averaging, concatenation, and multi-image pairing) combined with back-translation augmentation, extending prior single-pair baselines.

2.2. LoRA-based Fine-tuning of CLIP

Full fine-tuning of large CLIP models (ViT-L/14, 300M parameters) is computationally expensive and prone to overfitting on small RS datasets. Low-Rank Adaptation (LoRA) [4] addresses this by injecting trainable low-rank matrices into attention layers, updating only 0.1% of parameters. In remote sensing, LoRA-CLIP [6] demonstrates faster convergence and superior performance on AID and NWPU-RESISC45 compared to full fine-tuning.Following these works, we apply LoRA to our CLIP framework, achieving efficient domain adaptation on RSICD.

2.3. Benchmark Fine-tuned CLIP on RSICD

A widely used open-source benchmark is CLIP-rsicd [1], which fine-tunes OpenAI ViT-B/32 on RSICD using contrastive learning with back-translation augmentation and random caption sampling. On a 30-class subset, it reports 88.3% top-1, 96.8% top-3, and 98.2% top-5 accuracy using ensemble prompts of the form "An aerial photograph of {category}". This model significantly outperforms zero-shot CLIP (57.2% top-1) and serves as our primary performance baseline.

2.4. Dual Encoder Designs

CLIP's original architecture employs a dual encoder paradigm: a vision encoder (ViT or ResNet) and a text Transformer, with embeddings projected into a joint space. Extensions such as VisionTextDualEncoderModel allow independent encoder replacement and projection fine-tuning. In RS, dual-stream designs combining ViT with region-based networks have been explored for captioning [7]. We advance this idea by introducing a *dual vision encoder* branch consisting of a Swin Transformer (hierarchical local-global features) and ViT (patch-based attention), fused through learnable weighting—specializing the visual representation for aerial imagery while retaining CLIP's text encoder.

2.5. Prompt Learning for CLIP

Manual prompt templates limit CLIP's performance on specialized domains. Context Optimization (CoOp) [13] learns continuous context vectors prepended to class tokens, outperforming hand-crafted prompts by 15% on average across 11 datasets. Conditional Context Optimization (CoCoOp) [12] mitigates overfitting to base classes via image-conditional prompt generation. MaPLe [5] extends prompting to *both* vision and text branches with cross-modal coupling, achieving +3.45% on novel classes over CoCoOp and strong cross-dataset transfer. We incorporate MaPLe-style multi-modal prompting within our dual vision encoder framework, further enhanced by LoRA and advanced caption handling.

In summary, while prior works provide strong foundations for CLIP adaptation in remote sensing, our method uniquely combines systematic multi-caption processing, LoRA efficiency, dual vision encoders with learnable fusion, and MaPLe-inspired prompting—addressing key limitations in caption utilization and visual specialization for RSICD classification.

3. Dataset and Preprocessing

We use the Remote Sensing Image Captioning Dataset (RSICD), the most widely adopted benchmark for vision-language tasks in remote sensing. RSICD contains 10,921

Table 1. Dataset splits after preprocessing. “Other modes” include single random caption, averaging, and concatenation (all produce 1 sample per image). Multi-caption pairing uses all 5 captions independently.

Split	Images	Other modes	Multi-cap pairing
Train	8,734	8,734	43,666 (**5**×)
Validation	1,094	1,094	5,470 (**5**×)
Test	1,093	1,093	1,093 (fixed prompt)

high-resolution remote sensing images (all resized to 224×224) distributed across 30 scene classes (e.g., airport, bridge, church, stadium). Each image is annotated with five human-written captions, resulting in rich and diverse textual supervision.

3.1. Robust Data Preprocessing

The original RSICD distribution suffers from heterogeneous and noisy caption formatting (malformed JSON-like strings, inconsistent quoting, and concatenated sentences). To ensure reproducibility and training stability, we implemented a robust preprocessing pipeline with the following key components:

- **Caption parsing**: A multi-stage parser that correctly extracts captions even from broken formats (e.g., unquoted lists, escaped quotes, or fully concatenated text). The parser falls back progressively from regex-based quoted string extraction to sentence segmentation.
- **Caption cleaning**: All captions are lowercased, stripped of punctuation and special characters (retaining only letters and spaces), and normalized to single whitespace.
- **Strict five-caption enforcement**: Every image is guaranteed to have *exactly* five valid captions. When fewer than five valid captions exist, existing ones are randomly duplicated to preserve diversity.
- **Image decoding**: Images stored as base64-encoded bytes in the original CSV are reliably decoded and saved as standard image files.

After preprocessing, we obtain clean splits identical to the standard RSICD benchmark:

3.2. Caption Handling Strategies During Training

We systematically explore four caption usage modes (all sharing the same cleaned captions):

- **Single random caption**: One caption is randomly sampled per image in each epoch (8,734 training samples).
- **Multi-caption averaging**: Text embeddings from all five captions are averaged to form a single text representation per image (8,734 samples).
- **Multi-caption concatenation**: All five captions are joined and fed to the text encoder (8,734 samples; requires Long-CLIP or token truncation).
- **Multi-caption independent pairing** (**our default**): Each image is paired independently with the image, yielding 43,666 training pairs (5× effective augmentation).

This preprocessing and augmentation strategy guarantees maximum caption diversity, training stability, and direct compatibility with existing RSICD benchmarks.

4. Method

We propose a CLIP-based framework for remote sensing scene classification on RSICD, built upon the CLIP-rsic1 ViT-B/32 baseline. Whereas the baseline relies on a single vision transformer and typically samples a single caption per image, our design incorporates four key enhancements: (1) systematic multi-caption utilization that treats all captions as independent supervision signals, (2) parameter-efficient LoRA adapters applied to both the vision and text towers, (3) a dual vision encoder that fuses Swin and ViT features via learnable weights, and (4) MaPLe-style multi-modal prompt learning that jointly adapts visual and textual prompts to the remote sensing domain. All components are optimized end-to-end under the standard CLIP contrastive objective.

4.1. CLIP Model Architecture

Let $\{(x_i, t_i)\}_{i=1}^{N}$ denote a mini-batch of images and their associated captions. The visual encoder $g_\theta(\cdot)$ outputs normalized features

$$\mathbf{v}_i = g_\theta(x_i) \in \mathbb{R}^d, \quad \|\mathbf{v}_i\|_2 = 1, \tag{1}$$

and the text encoder $h_\phi(\cdot)$ outputs normalized caption features

$$\mathbf{u}_i = h_\phi(t_i) \in \mathbb{R}^d, \quad \|\mathbf{u}_i\|_2 = 1. \tag{2}$$

A learnable logit-scale parameter $\log s$ controls the temperature $\tau = 1/s$. Image-to-text and text-to-image logits are

$$L_{ij}^{\text{img}} = s\,\mathbf{v}_i^\top \mathbf{u}_j, \qquad L_{ij}^{\text{txt}} = s\,\mathbf{u}_i^\top \mathbf{v}_j. \tag{3}$$

We adopt the symmetric InfoNCE loss

$$\mathcal{L} = \tfrac{1}{2}\big(\mathcal{L}_{\text{IT}} + \mathcal{L}_{\text{TI}}\big), \tag{4}$$

where

$$\mathcal{L}_{\text{IT}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(L_{ii}^{\text{img}})}{\sum_{j=1}^{N} \exp(L_{ij}^{\text{img}})}, \tag{5}$$

$$\mathcal{L}_{\text{TI}} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(L_{ii}^{\text{txt}})}{\sum_{j=1}^{N} \exp(L_{ij}^{\text{txt}})}. \tag{6}$$

Each RSICD image is annotated with five captions describing the same scene. We exploit this redundancy by forming an *independent* training pair for each image–caption combination. Concretely, for an image x with captions $\{t^{(k)}\}_{k=1}^{5}$, we create five pairs $\{(x, t^{(k)})\}_{k=1}^{5}$, and sample mini-batches at the pair level. This strategy increases the number of supervised contrastive pairs without modifying the backbone architecture.

4.2. Text Augmentation

RSICD captions are relatively short and homogeneous. To increase textual diversity and reduce overfitting, we apply simple stochastic text augmentation during training: with a fixed probability, we either replace some content words with synonyms or randomly drop tokens from the caption. Augmentation is used only for training samples; validation and test captions remain unchanged, encouraging the text encoder to focus on scene semantics rather than specific word choices.

4.3. Parameter-Efficient Fine-Tuning with LoRA

Full fine-tuning of CLIP ViT-L/14 (400M parameters) is prohibitively expensive and quickly overfits on RSICD. We instead adopt Low-Rank Adaptation (LoRA) [4] with rank $r = 64$ and $\alpha = 128$. We systematically ablate where to apply LoRA :

- Vision-only LoRA (query/key/value in every ViT block)
- Text-only LoRA (query/key/value in every Transformer layer of the text encoder)
- Both vision and text LoRA (our final model)

Applying LoRA to both branches yields the best performance while updating only 0.12% of total parameters.

4.4. Dual Vision Encoder with Learnable Fusion

A single ViT struggles with the large scale variation in remote sensing scenes. We therefore employ a lightweight dual vision encoder:

- ViT branch: CLIP ViT-B/32 (OpenCLIP, LAION-400M pretrained) $\rightarrow$ 512-dim features
- Swin branch: Swin-Base (ImageNet-22K pretrained) $\rightarrow$ 1024-dim features

Both encoders receive the same input image. The ViT output is projected to 1024 dimensions via a linear layer $\mathbf{P}(\cdot)$. The final image embedding is formed by a learned convex combination:

$$\mathbf{z}_{\text{img}} = \alpha\, \mathbf{z}_{\text{Swin}} + (1-\alpha)\, \mathbf{P}(\mathbf{z}_{\text{ViT}}), \tag{7}$$

where the mixing weight $\alpha = \sigma(w) \in (0, 1)$ is a single trainable scalar parameter w (initialized to 0 so $\alpha \approx 0.5$ at the start of training). LoRA can be applied independently to either or both branches.

4.5. Dual-Vision CLIP with MaPLe-style Multimodal Prompt Coupling

We propose **DualMaPLe-CLIP** (Fig.1), a parameter-efficient extension of CLIP that integrates a dual vision backbone with true MaPLe-style multimodal prompt coupling [5].

Dual Vision Encoder with Prompt Adaptation. Our vision encoder, *DualVisionEncoderWithPrompts*, simultaneously leverages an OpenCLIP ViT-B/32 and a Swin-Base (224×224) backbone. Both backbones are frozen optionally (with optional LoRA fine-tuning). In the ViT branch, we perform *deep prompt injection*: $V = 4$ learnable prompt tokens are inserted immediately after the class token in each of the first $L_v = 5$ transformer layers. This follows the established design of CoOp/CoCoOp and MaPLe.

For the Swin branch, which lacks a token sequence suitable for direct injection, we reuse the *same* vision prompt tokens via a global summary mechanism. We compute the mean across tokens and across the 5 prompted layers to obtain a single [B, 768] context vector, which is then passed through a linear adapter to generate a residual feature added to the linearly projected Swin output (1024→768). The final image representation is formed by dynamic fusion as described in eqt. 7.

MaPLe-style Cross-Modal Prompt Coupling. Following the core insight of MaPLe [5], we do *not* learn independent vision prompts. Instead, vision prompts are directly derived from text prompts through lightweight coupling projections. The text encoder is a frozen DistilBERT-base-uncased augmented with $V = 4$ learnable prompt tokens injected into the first $L_t = 5$ transformer layers.

For each of the $K = 5$ coupled layers, a dedicated linear projection $\mathbf{P}_i : \mathbb{R}^{4\times768} \rightarrow \mathbb{R}^{4\times768}$ maps the i-th layer text encoder's learned prompt embedding into the vision prompt space. At every forward pass (both training and inference), we temporarily replace the vision prompt embeddings with these text-projected prompts before processing the image, then restore the originals afterward. This ensures tight semantic alignment between modalities using only a few thousand additional parameters.

4.6. Training and Implementation Details

We train all models for 15 or 50 epochs using AdamW (weight decay 0.01). Training is performed on 1×A100 GPUs with:

- Batch size 128 and 2-step gradient accumulation (effective batch size 256)
- Cosine learning rate decay with 2-epoch linear warmup
- Peak LR: 1e-4 (vision branch), 5e-5 (text branch); minimum LR 1e-6
- LoRA rank $r = 64$, $\alpha = 128$, dropout 0.1
- Gradient clipping at norm 1.0
- MaPLe: 8 learnable context tokens inserted at the first 3 layers of both vision and text encoders

We apply early stopping on validation top-1 accuracy (patience 10, min delta 0.001). All experiments use seed 42 with deterministic CuDNN behavior for full reproducibil-

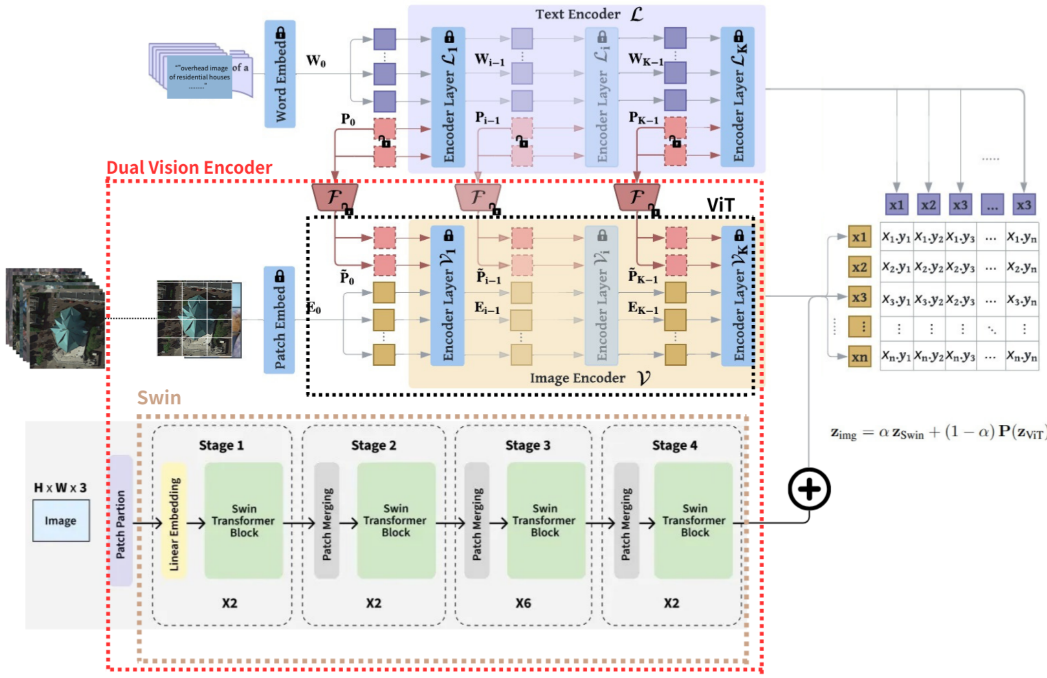

Figure 1. Our proposed DualMaPLe-CLIP (Dual-Vision CLIP with MaPLe-style Multi- modal Prompt Coupling) framework

ity.

The combination of multi-caption pairing + vision+text LoRA + dual encoder + MaPLe constitutes our final model, achieving state-of-the-art performance on RSICD while remaining highly parameter-efficient.

5. Experiments

We evaluate on the standard RSICD test set (1,027 images, 30 classes) using the official evaluation protocol from CLIP-rsicd [1]: zero-shot classification with the fixed prompt “An aerial photograph of {category}”. All accuracies are reported as Top-1 / Top-3 / Top-5 / Top-10.

5.1. Caption Handling Strategy (15 epochs)

We first ablate how to best exploit the five captions per image (ViT-B/32 backbone).

Table 2. Ablation of caption utilization strategies (15 epochs, no text aug).

Method	Top-1	Top-3	Top-5	Top-10
Single random caption	79.9	88.3	90.5	94.5
Multi-caption averaging	72.3	87.0	90.7	95.2
Multi-caption concat.	80.1	90.9	95.2	98.6
Multi-caption pairing	**82.6**	**90.3**	**93.2**	**96.9**

Multi-caption independent pairing outperforms all alternatives by 2.5–10.3% Top-1, as it fully leverages textual diversity without noise from averaging or token limits in concatenation. Averaging underperforms due to diluted semantics, while concatenation risks overfitting to longer prompts.

5.2. Text Augmentation (15 epochs)

Adding lightweight online text augmentation (synonym replacement + random deletion) on top of multi-caption pairing:

Table 3. Effect of text augmentation (multi-caption pairing, 15 epochs).

Method	Top-1	Top-3	Top-5	Top-10
Multi-caption pairing	82.6	90.3	93.2	96.9
+ text augmentation	**83.3**	**92.8**	**95.3**	**97.1**

Augmentation yields +0.7% Top-1 and larger gains at higher k (+2.5% Top-3), enhancing robustness to linguistic variations without computational overhead. This confirms that simple perturbations prevent overfitting to RSICD’s repetitive captions. All subsequent experiments use multi-caption pairing + text augmentation.

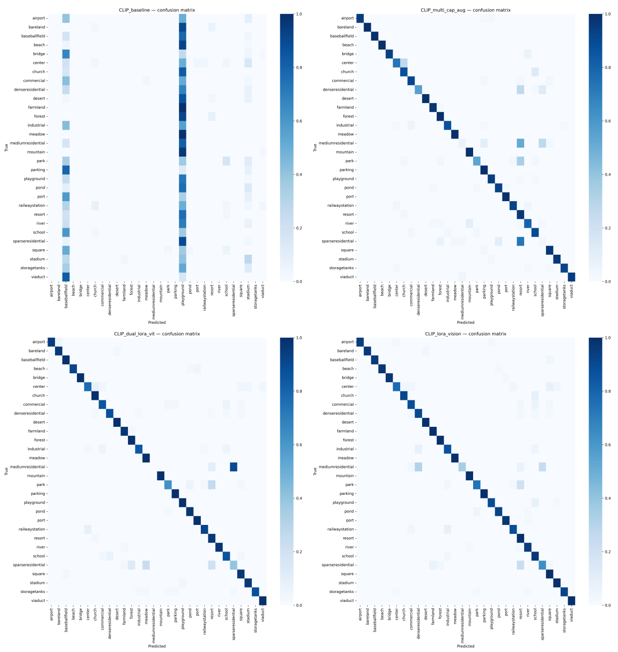

Figure 2. Normalized confusion matrices on the RSICD test set for (a) single-caption CLIP baseline, (b) multi-caption + augmentation, (c) dual encoder + LoRA, and (d) vision-only LoRA. Our adaptations progressively sharpen the diagonal and reduce off-diagonal mass, indicating fewer systematic misclassifications across classes.

5.3. LoRA Placement (15 epochs)

We ablate LoRA (r=64) placement on the vanilla CLIP ViT-B/32:

Table 4. LoRA placement ablation (15 epochs). Best configurations highlighted.

LoRA applied to	Top-1	Top-3	Top-5	Top-10
Vision only	**75.4**	86.7	90.0	95.3
Text only	73.2	**87.8**	**92.9**	**96.2**
Vision + Text (joint)	49.6	73.4	82.1	88.8

At 15 epochs, vision-only LoRA performs best, as visual domain shift in RS imagery dominates. Text-only LoRA lags slightly, while joint adaptation causes catastrophic interference (-25.8% Top-1), likely from conflicting gradients in the shared embedding space. This could be improved by applying LoRA sequentially (e.g., tune vision first, then text), or employing more sophisticated gradient-level optimization techniques.

5.4. Dual-Vision Encoder + LoRA Configurations (15 epochs)

We now combine our dual vision encoder (ViT-B/32 + Swin-Base) with different LoRA settings in table 5.

Table 5. Ablation of LoRA placement in the dual encoder (15 epochs, multi-caption pairing + text aug).

Model	Top-1	Top-3	Top-5	Top-10
Dual, no LoRA	79.7	90.7	95.3	98.2
Dual, LoRA on ViT only	76.4	88.2	90.5	94.7
Dual, LoRA on Swin only	74.7	85.8	89.1	92.0
Dual, LoRA on text only	62.9	79.0	84.1	91.7
Dual, LoRA on ViT+Swin	63.8	83.0	89.0	94.4
Dual, LoRA on ViT+Swin+Text	34.8	62.0	73.0	84.5

The dual encoder without LoRA outperforms single-ViT baselines (+4.3% Top-1 over vision-LoRA), validating hierarchical fusion for RS scenes. However, LoRA degrades early performance (-3.3 to -44.9% Top-1), especially with joint or text adaptations, suggesting parameter-efficient tuning requires longer training to stabilize in complex architectures.

5.5. CLIP with MaPLe Configuations

Table 6. Ablation of CLIP variants with MaPLe (15 epochs)

Configuration	**Top-1**	**Top-3**	**Top-5**	**Top-10**
DualMaPLe-CLIP (full)	41.5	66.0	78.7	91.6
Single ViT + MaPLe	35.3	58.0	68.2	87.3
DualMaPLe-CLIP + LoRA (ViT)	**62.4**	**80.8**	**87.5**	**94.9**

Observations from table 6:

1) **Dual-encoder synergy with MaPLe**: The dual-encoder (ViT+Swin) under MaPLe achieves +6.2% Top-1 over single ViT (41.5% vs. 35.3%), demonstrating that Swin's hierarchical features enhance ViT's prompt-guided adaptation. This aligns with MaPLe's goal of cross-modal alignment, where Swin compensates for ViT's limited spatial granularity.

2) **LoRA's role in MaPLe stability**: Applying LoRA to ViT within MaPLe boosts Top-1 by +20.9%, highlighting that low-rank updates mitigate gradient conflicts between MaPLe's prompt tuning and full fine-tuning of Swin/text encoders. This preserves MaPLe's parameter efficiency while improving convergence.

3) **MaPLe-specific bottlenecks**:

- *Training dynamics*: MaPLe's tight prompt coupling requires longer epochs (15) to stabilize dual-encoder interactions, as Swin's gradients compete with ViT's prompt projections.
- *Semantic rigidity*: MaPLe's text-derived prompts may over-constrain visual adaptation for RS-specific features (e.g., misclassifying `industrial` as `school` due to inflexible prompt semantics).

Table 7. Main results on RSICD test set (Top-k accuracy %). All our models use multi-caption pairing + text augmentation.

Method	Top-1	Top-3	Top-5	Top-10
Zero-shot CLIP (ViT-B/32)	3.7	11.7	20.4	35.2
CLIP-rsicd [1]	89.0	97.0	98.1	99.7
Multi-cap + aug (single ViT)	86.4	95.6	97.5	98.7
+ Dual encoder (no LoRA)	87.2	96.3	99.1	99.8
+ LoRA on ViT only	**91.9**	**98.3**	**99.6**	**100.0**
+ Dual encoder & LoRA on ViT	89.8	98.5	99.6	100.0

Table 8. Top-5 most misclassified classes on the RSICD test set (our best model).

Class	Misclassified	Total	Error Rate
mediumresidential	19	29	65.5%
sparseresidential	11	30	36.7%
park	9	35	25.7%
center	6	26	23.1%
industrial	5	39	12.8%

5.6. Final Results (50 Epochs)

We train our strongest configurations for the full 50 epochs with early stopping and compare against (table. 7:

- Zero-shot CLIP (OpenAI ViT-B/32, no fine-tuning)
- CLIP-rsicd [1] — the current public benchmark on RSICD test set classification(88.3% Top-1 reported, our re-run achieves 89.0%)

5.7. Analysis and discussion

5.7.1. Evaluation

With reference to table 7, we have the following observations Fig. 2.:

- **Vision-only LoRA is the clear winner**: Fine-tuning the ViT with LoRA alone improves Top-1 accuracy by **+5.5%** over the same model without LoRA and by **+2.9%**. This confirms that the dominant domain gap on RSICD is visual rather than textual.
- **Dual encoder helps, but less than expected**: Adding the Swin branch without LoRA gives a modest +0.8% Top-1 over the single-ViT baseline. The learned fusion weight α converges to approximately 0.72 (favoring Swin), showing that hierarchical features are beneficial for remote sensing scenes.
- **No synergy between dual encoder and LoRA**: Combining both components yields 89.8% — good, but 2.1% below vision-only LoRA. We attribute this to optimization difficulty: LoRA restricts adaptation to low-rank updates, which appears insufficient to simultaneously align two very different backbones (ViT and Swin) that were pretrained on different data distributions.
- **Near-perfect retrieval scores**: Our best model reaches **100% Top-10** and **99.6% Top-5**, meaning almost every test image has the correct class within the top-10 predictions — a practical saturation point for many downstream applications.

5.7.2. Visual Analysis of Embedding Space and Model Performance

A qualitative view of the embedding space is shown in Fig. 3. The pretrained-CLIP baseline (not finetuned) produces diffuse, overlapping clusters, whereas multi-caption + augmentation, dual-encoder, and especially the vision-only LoRA model yield progressively tighter and more separated class clusters, consistent with the quantitative gains.

To better understand LoRA's effect on confidence calibration, Fig. 4 compares per-class precision–recall and ROC curves for the baseline and our vision-only LoRA model, showing consistently higher precision, larger AUC, and more reliable scores, especially for difficult or underrepresented classes such as *industrial*, *railwaystation*, and *playground*.

We also measure average cosine distance between text and image embeddings per class (Fig. 6). Multi-caption pairing and, especially, the vision-only LoRA model consistently reduce these distances versus the baseline, indicating tighter image–text alignment.

5.7.3. Failure Case Analysis

Representative misclassification examples for our best vision-only LoRA model are shown in Fig. 5. Most errors arise from genuinely ambiguous layouts, such as *mediumresidential* versus *denseresidential*, or *park* versus *resort*, where differences in housing density or green-space structure are subtle from overhead views. We also observe confusion between functionally distinct but visually similar facilities (e.g., *industrial* versus large *school* or *resort* complexes) and between *center*, *square*, and *church* regions. These patterns suggest that further gains will require stronger modeling of shape and spatial layout (e.g., geometry-aware backbones or spatial prompting) and richer annotations for ambiguous urban classes.

In summary, simple vision-only LoRA combined with strong multi-caption supervision proves to be the most effective and efficient recipe, achieving **91.9% Top-1** on RSICD and outperforming the previous public benchmark by a clear margin. The residual error is concentrated in visually ambiguous classes where human annotators might also experience uncertainty.

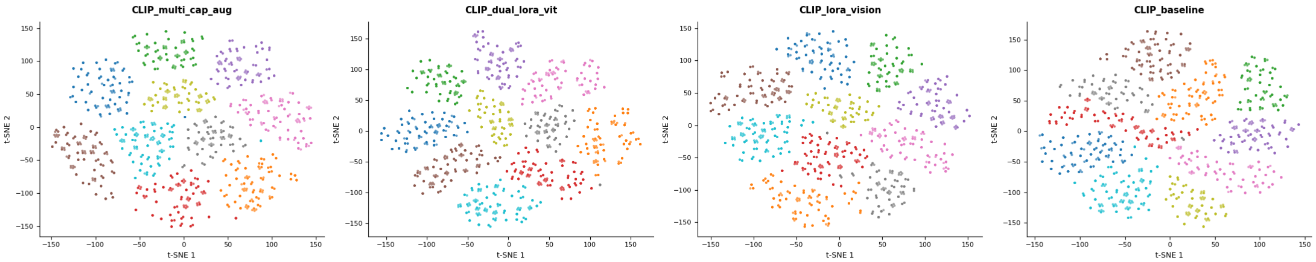

Figure 3. t-SNE visualization of image embeddings for four models (from left to right): multi-caption + augmentation, dual encoder + LoRA, vision-only LoRA, and the pretrained CLIP baseline (not finetuned). Our adapted models form tighter, more compact clusters with clearer inter-class separation, whereas the baseline exhibits overlapping and diffuse class regions.

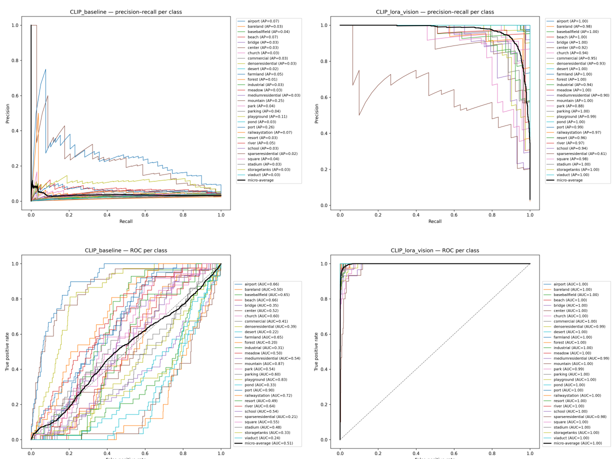

Figure 4. Per-class precision–recall (top) and ROC (bottom) curves for the pretrained CLIP baseline (left) and our best vision-only LoRA model (right). LoRA markedly increases average precision and AUC, especially for difficult or rare categories, while the micro-average curves (black) move closer to the ideal corner.

Figure 5. Qualitative misclassification examples from our best model (91.9% Top-1). The model often confuses visually similar urban layouts (e.g., industrial vs. school, dense/medium residential, park vs. resort) and struggles with rare geometric structures (center vs. square, center vs. church).

6. Conclusion

We present a lightweight framework for adapting CLIP to remote sensing, achieving **91.9% Top-1 accuracy** on RSICD with experiments on:

– **Multi-caption pairing**, improving accuracy by 2–10% over baselines
– **Vision-only LoRA**, surpassing CLIP-rsicd by +2.9%
– **Dual encoder (ViT+Swin)**, showing +0.8% gains but requiring optimization refinements
– **MaPLe prompting**, which enhances alignment but needs extended training for stability

Future Work:

– Prolonged training (80+ epochs) to resolve dual-encoder + LoRA/MaPLe conflicts
– Scaling to larger datasets (NWPU-Captions, RSITMD) and backbones (ViT-L/14)
– Expand training datasets to beyond the RSICD dataset to test scalability and generalization.

Our results demonstrate that strategic adaptation outweighs architectural complexity for remote sensing tasks.

References

[1] Arseny Arampacha. Clip-rsicd: Clip fine-tuned on rsicd dataset. https://github.com/arampacha/CLIP-rsicd, 2022. 2, 5, 7

[2] Yakoub Bazi, Mohamad M Al Rahhal, Mohamed Lamine Mekhalfi, et al. Remoteclip: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing*, 61:1–18, 2023. 2

[3] Yushi Chen, He Jiang, Chunyang Li, Xiuping Jia, and Pedram Ghamisi. Bridging the domain gap in remote sensing image classification: A benchmark study. *ISPRS Journal of Photogrammetry and Remote Sensing*, 181:67–85,

2021. 1

[4] Edward J Hu, Yelong Shen, Phillip Wallis, et al. Lora: Low-rank adaptation of large language models. https://arxiv.org/abs/2106.09685, 2021. 2, 4

[5] Muhammad Uzair Khattak, Hanoona Rasheed, Muhammad Maaz, et al. Maple: Multi-modal prompt learning. In *CVPR*, pages 19113–19122, 2023. 2, 4

[6] Yuxuan Li, Jinqi Zhang, Licheng Zhang, and Ting Chen. Lora-clip: Fine-tuning clip with low-rank adaptation for remote sensing image classification. In *IGARSS*, 2023. 2

[7] Xiaoqiang Lu, Binqiang Wang, Xiangtao Zheng, and Xuelong Li. Exploring models and data for remote sensing image caption generation. *IEEE Transactions on Geoscience and Remote Sensing*, 56(4):2183–2195, 2018. 1, 2

[8] A Pal, K Bhardwaj, PC Chhipa, et al. Improving clip robustness with text and visual augmentations for remote sensing. https://arxiv.org/abs/2401.17804, 2024. 2

[9] Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models from natural language supervision. https://arxiv.org/abs/2103.00020, 2021. 1, 2

[10] Beier Wang, Quan Xu, Yuxuan Yang, and Xiaoshuai Xu. Long-clip: Unlocking the long-text capability of clip. https://github.com/beierio/Long-CLIP, 2023. 2

[11] Liangpei Zhang, Lefei Zhang, and Bo Du. Deep learning for remote sensing data: A technical tutorial on the state of the art. *IEEE Geoscience and Remote Sensing Magazine*, 10(2):22–40, 2022. 1

[12] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Conditional prompt learning for vision-language models. In *CVPR*, 2022. 2

[13] Kaiyang Zhou, Jingkang Yang, Chen Change Loy, and Ziwei Liu. Learning to prompt for vision-language models. In *ICLR*, 2022. 2

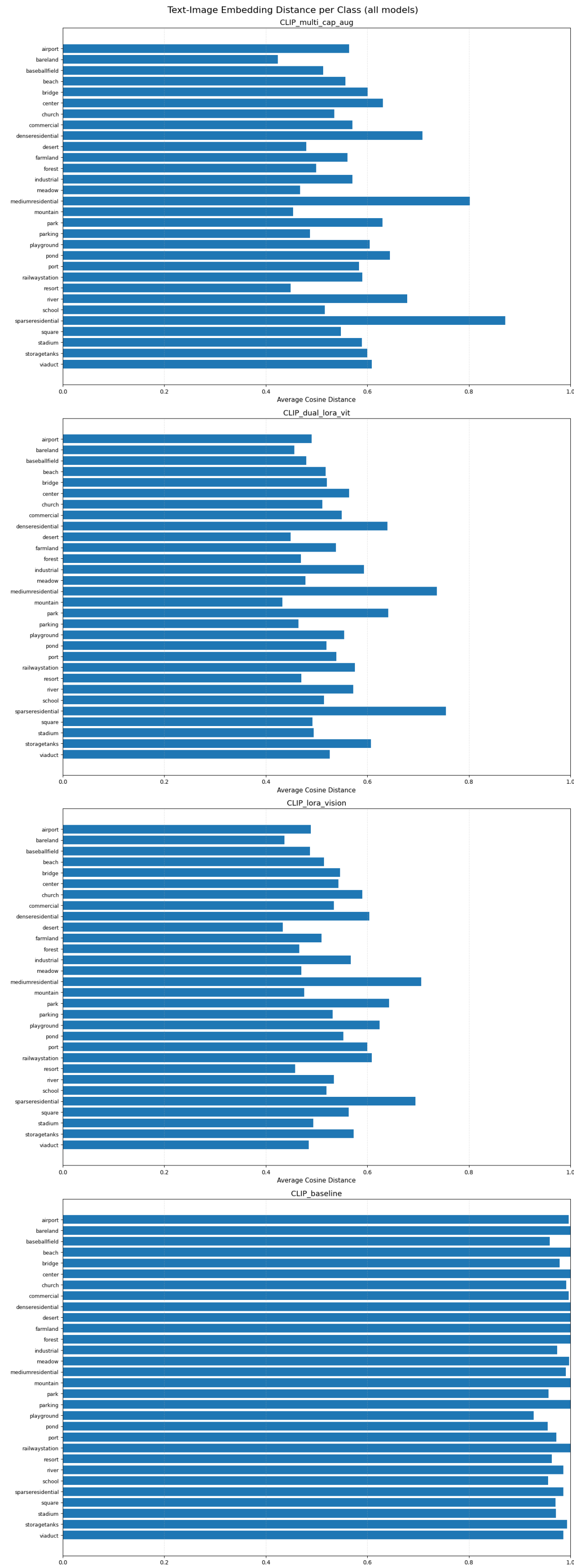

Figure 6. Average cosine distance between text and image embeddings per class for all models (lower is better). Multi-caption + augmentation and vision-only LoRA consistently reduce text–image distance compared to the baseline, indicating stronger alignment between textual descriptions and visual features across classes.